

Relation Algebra

§15. Dynamic Programming

§15.1. Theory

$h : FB \rightarrow B$ a map, $T : FA \rightarrow A$ an F -algebra, $R : B \rightarrow B$.

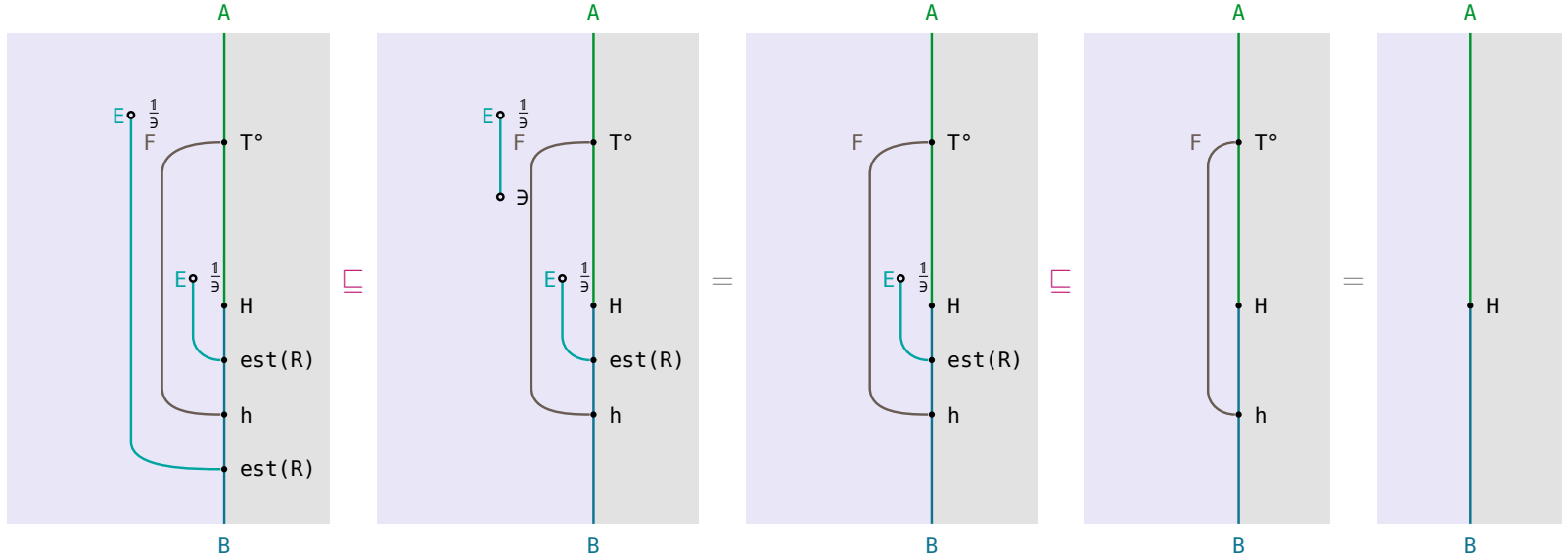
(15.1a)

$H \triangleq (T)^\circ \circ (h) : A \rightarrow B$, $M \triangleq \frac{H}{\exists} \text{ est}(R)$ the problem to be solved, $(\mu X : G(X))$ as in $\triangleq \mu$.

$$T^\circ F(H)h = H \Rightarrow \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{ est}(R))h) \text{ est}(R) \sqsubseteq H$$

(15.1b)

taking the input apart every way T allows $(\frac{T^\circ}{\exists})$, solving each part by M and keeping an optimum $(P(F(M)h) \text{ est}(R))$ returns only what H returns



$$\begin{aligned} & A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{P(F(\frac{H}{\exists} \text{ est}(R))h)} EB \xrightarrow{\text{est}(R)} B \\ \sqsubseteq & A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\exists} FA \xrightarrow{F(\frac{H}{\exists} \text{ est}(R))} FB \xrightarrow{h} B \\ = & A \xrightarrow{T} FA \xrightarrow{F(\frac{H}{\exists} \text{ est}(R))} FB \xrightarrow{h} B \\ \sqsubseteq & A \xrightarrow{T} FA \xrightarrow{F(H)} FB \xrightarrow{h} B \\ = & A \xrightarrow{H} B \end{aligned}$$

$$(9.4) P(X) \text{ est}(R) \sqsubseteq \exists X \text{ at } X \triangleq F(\frac{H}{\exists} \text{ est}(R))h \text{ — est-710}$$

$$\frac{T^\circ}{\exists} \exists = T^\circ \text{ — pow-laws}$$

$$M \triangleq \frac{H}{\exists} \text{ est}(R) \sqsubseteq \frac{H}{\exists} \exists = H \text{ — est-up}$$

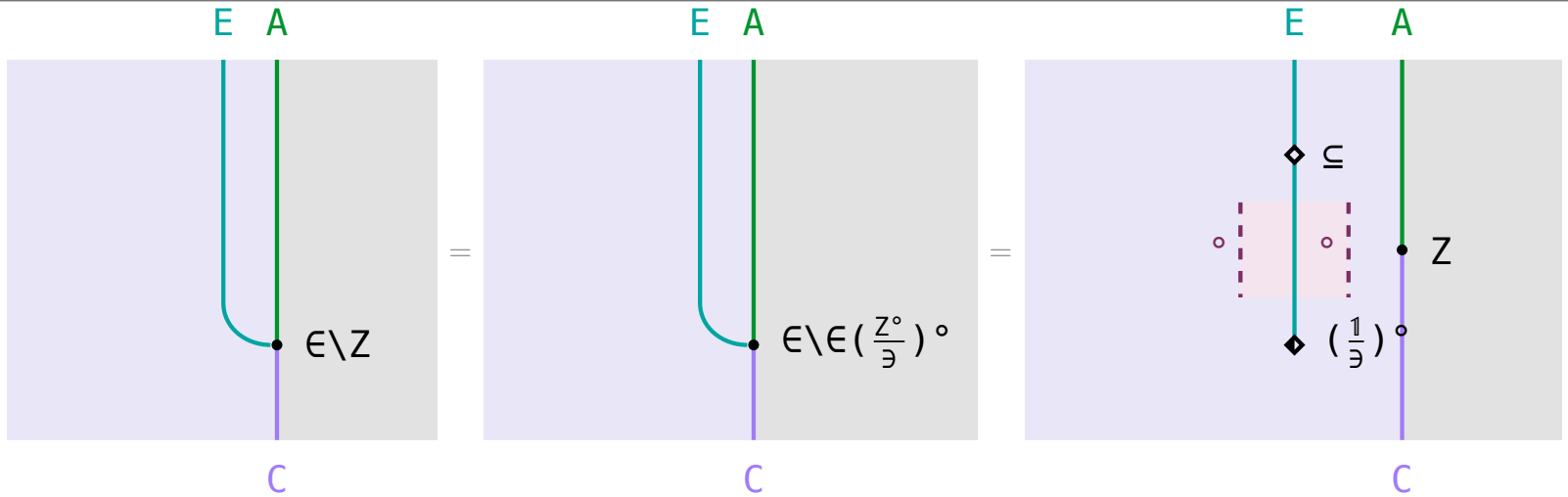
$$T^\circ F(H)h = H: H \triangleq (T)^\circ \circ (h) \text{ and hylo-fix}$$

§15.1.1. $\epsilon \setminus$ as a composite

$$\epsilon \setminus Z = \epsilon \setminus (\frac{Z^\circ}{\exists})^\circ$$

(15.1.1a)

xs is related to c by $\epsilon \setminus Z$ exactly when $xs \sqsubseteq \Lambda(Z^\circ)(c)$, the set of every a with $a Z c$



$$\begin{aligned} & EA \xrightarrow{\epsilon \setminus Z} C \\ = & EA \xrightarrow{\epsilon \setminus \epsilon \setminus (\frac{Z^\circ}{\exists})^\circ} C \\ = & EA \xrightarrow{\sqsubseteq} E(Z^\circ) \xrightarrow{EC} \frac{1}{\exists} \xrightarrow{\quad} C \end{aligned}$$

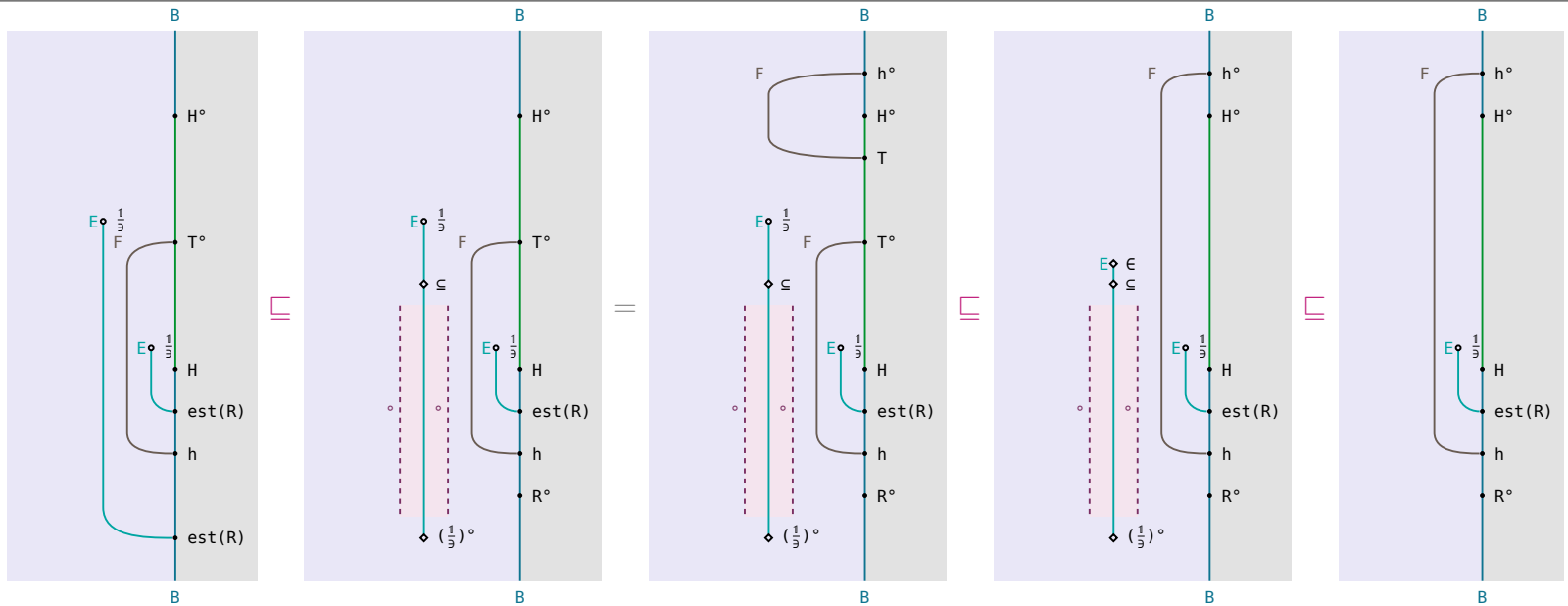
$$Z = \epsilon \setminus \Lambda(Z^\circ)^\circ: \text{the converse of } \Lambda(Z^\circ) \exists = Z^\circ \text{ — pow-laws}$$

$$X \setminus (Yf^\circ) = (X \setminus Y) f^\circ \text{ for a map } f \text{ (not a tabulated row), at the map } f \triangleq \Lambda(Z^\circ); \sqsubseteq \triangleq \epsilon \setminus \epsilon$$

(15.1.1b)

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H)h = H \implies H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R))h) \text{est}(R) \subseteq R^\circ$$

for every b that H returns from an input, the step $\frac{T^\circ}{\exists} P(F(M)h) \text{est}(R)$ returns from that input only b' with R relating b' to b



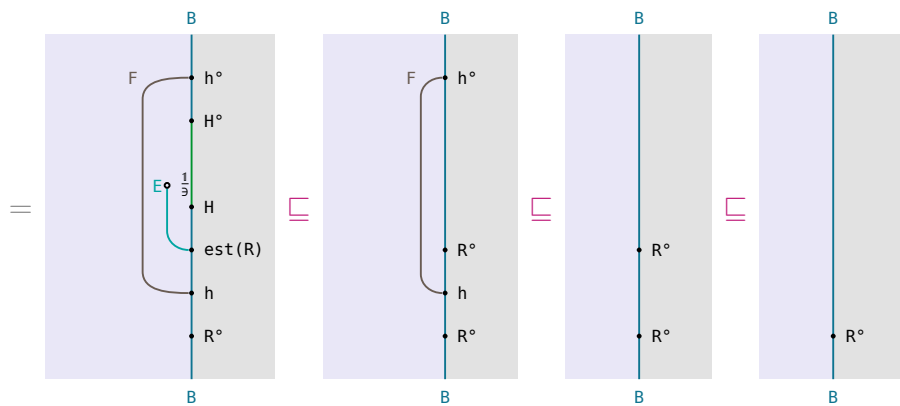
$$\begin{aligned} & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{P(F(\frac{H}{\exists} \text{est}(R))h)} EB \xrightarrow{\text{est}(R)} B \\ \sqsubseteq & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))h R^\circ} B \\ = & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{T} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))h R^\circ} B \\ \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{\epsilon} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))h R^\circ} B \\ \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} R \xrightarrow{} B \end{aligned}$$

$$(9.4) P(X) \text{est}(R) \subseteq \epsilon \setminus (X R^\circ) \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ — est-710}$$

$$H^\circ = h^\circ F(H^\circ)T: H \triangleq (T)^\circ(h) \text{ and hylo-fix}$$

$$T \frac{T^\circ}{\exists} \subseteq \epsilon: \frac{T^\circ}{\exists} \text{ a map, } \frac{T^\circ}{\exists} \exists = T^\circ \text{ — pow-laws}$$

$$\epsilon(\epsilon \setminus Y) \subseteq Y \text{ at } Y \triangleq F(M)h R^\circ \text{ — division}$$



$$\begin{aligned} = & B \xrightarrow{h} F(H^\circ \frac{H}{\exists} \text{est}(R)) \xrightarrow{FB} h \xrightarrow{R} B \\ \sqsubseteq & B \xrightarrow{h} F(R^\circ) \xrightarrow{FB} h \xrightarrow{R} B \\ \sqsubseteq & B \xrightarrow{R} R \xrightarrow{} B \\ \sqsubseteq & B \xrightarrow{R} B \end{aligned}$$

$$F(H^\circ)F(M) = F(H^\circ M): F \text{ a functor}$$

$$H^\circ M \subseteq R^\circ: M \triangleq \frac{H}{\exists} \text{est}(R) \text{ — est-up}$$

$$h^\circ F(R^\circ)h \subseteq R^\circ: h \text{ monotonic on } R^\circ$$

$$R^\circ R^\circ \subseteq R^\circ: R \text{ transitive}$$

(15.1.1c)

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge QF(H)h \subseteq F(H)h R \implies (\mu X : \frac{T^\circ}{\exists} \text{thin}(Q)P(F(X)h) \text{est}(R)) \subseteq \frac{H}{\exists} \text{est}(R)$$

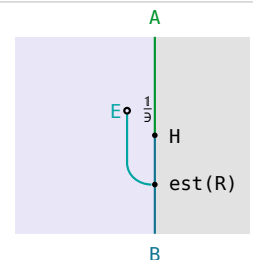
an optimum over everything H returns is reached by taking the input apart every way T allows, dropping the parts that can never win, solving each of the rest and keeping one optimum

circuit

$$A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(H)} EB \xrightarrow{\text{est}(R)} B$$

the problem to be solved, $H \triangleq (T)^\circ(h) \text{ — } \triangleq H$

Hinze–Marsden



circuit	Hinze–Marsden
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EFA \rightarrow \boxed{P(F(X)h)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B$ $\sqsubseteq$ h monotonic on R and Q a preorder with $QF(H)h \subseteq F(H)hR$; $\text{thin}(Q)$ as in thin-laws. Theorem 9.1 is this with the thinning step dropped — $1 \sqsubseteq \text{thin}(Q)$. Knaster–Tarski leaves (9.1): the body at M is $\sqsubseteq M$; $M = Hn(H^\circ \setminus R^\circ)$ splits that into (9.2) and (9.3) below — est-up. The fixed point is unique and entire when T° followed by F's membership relation is inductive, $\frac{T^\circ}{\exists}$ finite and non-empty, R connected	
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EFA \rightarrow \boxed{P(F(\frac{H}{\exists} \text{est}(R))h)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B$ (9.2)	
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EFA \rightarrow \boxed{\exists} \rightarrow FA \rightarrow \boxed{F(\frac{H}{\exists} \text{est}(R))} \rightarrow FB \rightarrow \boxed{h} \rightarrow B$ (9.4) $P(X)\text{est}(R) \sqsubseteq \exists X$ at $X \triangleq F(\frac{H}{\exists} \text{est}(R))h$ — est-710	
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow EFA \rightarrow \boxed{\exists} \rightarrow FA \rightarrow \boxed{F(\frac{H}{\exists} \text{est}(R))} \rightarrow FB \rightarrow \boxed{h} \rightarrow B$ $\text{thin}(Q) \exists \exists \exists$ — thin-laws	
$A \rightarrow \boxed{T} \rightarrow FA \rightarrow \boxed{F(\frac{H}{\exists} \text{est}(R))} \rightarrow FB \rightarrow \boxed{h} \rightarrow B$ $\frac{T^\circ}{\exists} \exists = T^\circ$ — pow-laws	

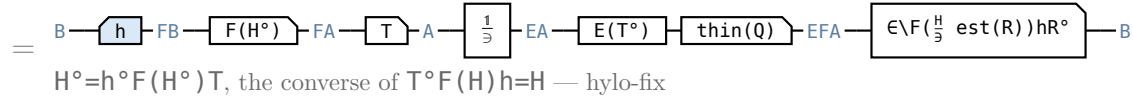
(15.1.1d)

circuit	Hinze–Marsden
$\begin{array}{c} \text{A} \text{---} \boxed{\text{T}} \text{---} \text{FA} \text{---} \boxed{\text{F(H)}} \text{---} \text{FB} \text{---} \boxed{\text{h}} \text{---} \text{B} \\ \text{M} \sqsubseteq \frac{\text{H}}{\text{F}} \text{---} \text{H} \text{---} \text{est-up} \end{array}$	
$\begin{array}{c} \text{A} \text{---} \boxed{\text{H}} \text{---} \text{B} \\ \text{T}^\circ \text{F(H)} \text{h} = \text{H}: \text{H} \triangleq (\text{T})^\circ \circ (\text{h}) \text{ and hylo-fix} \end{array}$	

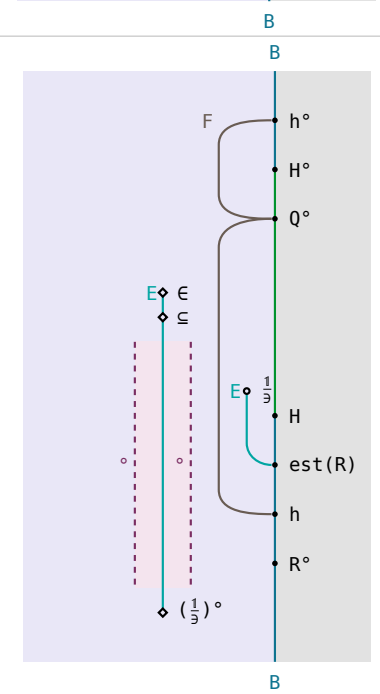
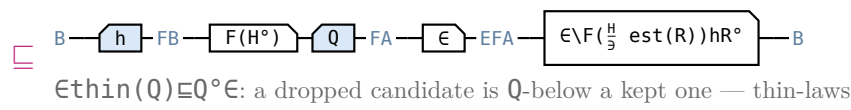
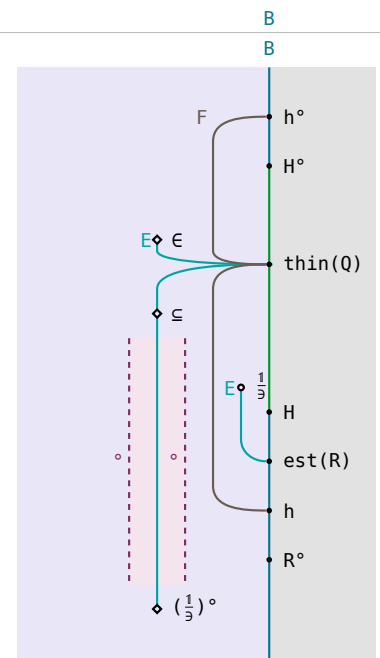
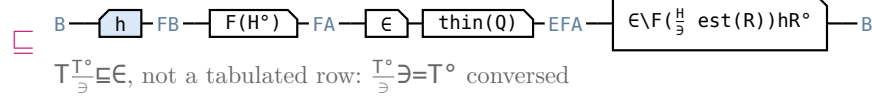
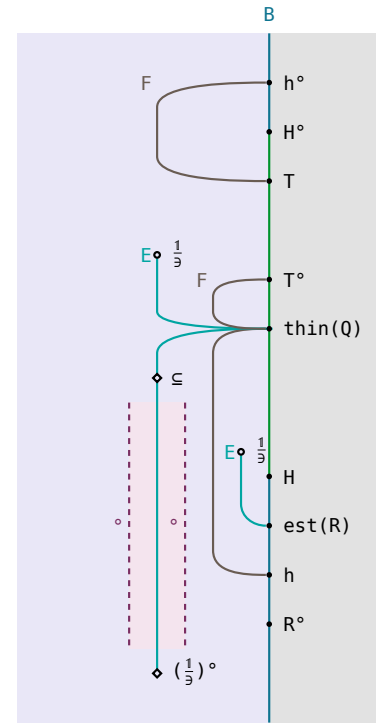
(15.1.1e)

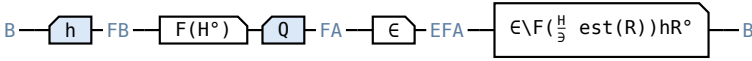
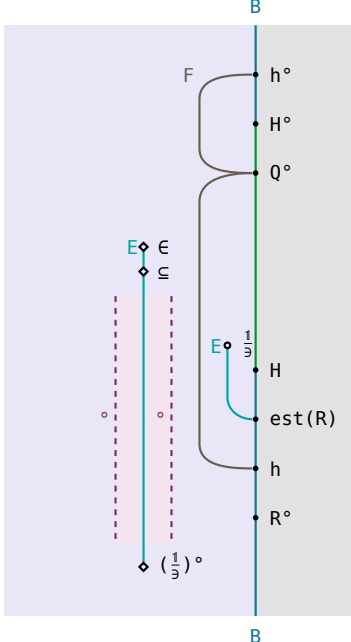
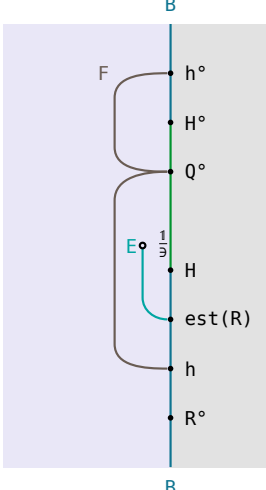
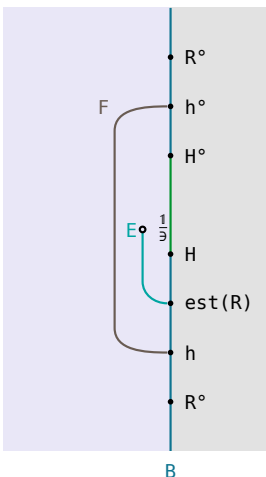
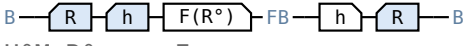
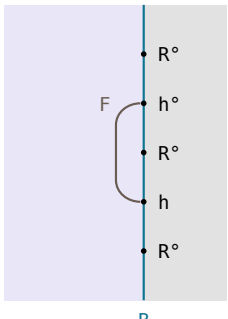
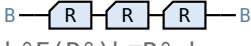
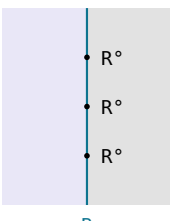
$\text{PreservesRecip(F)} \wedge \text{Map(h)} \wedge \text{MonotonicAlg(h, R}^\circ) \wedge \text{R}^\circ \text{R}^\circ \sqsubseteq \text{R}^\circ \wedge \text{T}^\circ \text{F(H)} \text{h} = \text{H} \wedge \text{QF(H)} \text{h} \sqsubseteq \text{F(H)} \text{hR} \implies \text{H}^\circ \frac{\text{T}^\circ}{\text{F}} \text{thin(Q)} \text{P(F(H)} \frac{\text{H}}{\text{F}} \text{est(R))h)est(R)} \sqsubseteq \text{R}^\circ$	
H° followed by the body at M is ER° : an answer of the body is never worse than an answer of H to the same input	
circuit	Hinze–Marsden
$\text{B} \text{---} \boxed{\text{H}} \text{---} \text{A} \text{---} \boxed{\frac{1}{\text{F}}} \text{---} \text{EA} \text{---} \boxed{\text{E(T}^\circ\text{)}} \text{---} \boxed{\text{thin(Q)}} \text{---} \text{EFA} \text{---} \boxed{\text{P(F(H)} \frac{\text{H}}{\text{F}} \text{est(R))h)}} \text{---} \text{EB} \text{---} \boxed{\text{est(R)}} \text{---} \text{B}$	
$\begin{array}{c} \text{B} \text{---} \boxed{\text{H}} \text{---} \text{A} \text{---} \boxed{\frac{1}{\text{F}}} \text{---} \text{EA} \text{---} \boxed{\text{E(T}^\circ\text{)}} \text{---} \boxed{\text{thin(Q)}} \text{---} \text{EFA} \text{---} \boxed{\text{E} \setminus \text{F(H)} \frac{\text{H}}{\text{F}} \text{est(R))hR}^\circ} \text{---} \text{B} \\ (9.4) \text{P(X)} \text{est(R)} \sqsubseteq \text{E} \setminus \text{X} \text{R}^\circ \text{ at } \text{X} \triangleq \text{F(H)} \frac{\text{H}}{\text{F}} \text{est(R))h} \text{---} \text{est-710} \end{array}$	

circuit



Hinze–Marsden



circuit	Hinze–Marsden
	
 $E(E \setminus Y) \sqsubseteq Y$, not a tabulated row: division cancels	
 $QF(H) h \sqsubseteq F(H) h R$ conversed — the hypothesis on Q	
 $H^\circ M \sqsubseteq R^\circ$ under F — est-up	
 $h^\circ F(R^\circ) h \sqsubseteq R^\circ$: h monotonic on R, shunted — the hypothesis on h	

circuit	Hinze–Marsden
$\begin{array}{c} B \text{---} \boxed{R} \text{---} B \\ \sqsubseteq \\ R \text{ transitive, twice} \end{array}$	

$$V_2 \quad V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \text{est}(R) = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{thin}(Q_1) P(U_1) \text{est}(R) \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{thin}(Q_2) P(U_2) \text{est}(R)$$

(15.1.1g)

when V_1 and V_2 have disjoint ranges, the body over $[V_1, V_2]$ runs the V_1 problem on inputs V_1 reaches and the V_2 problem on inputs V_2 reaches

circuit	Hinze–Marsden
$A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B$	
$\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ off $\text{ran } V_1 \cup \text{ran } V_2$ the set is empty and $\text{est}(R)$ of it is nothing; not a tabulated row	
$\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_1^\circ) \text{---} E\alpha \text{---} P(l) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_2^\circ) \text{---} E\beta \text{---} P(r) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ Exercise 9.5: on $\text{ran } V_1$, $\frac{[V_1, V_2]^\circ}{\exists} = \frac{V_1^\circ}{\exists} P(\text{in } l)$, as V_1, V_2 have disjoint ranges	
$\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_1^\circ) \text{---} \text{thin}(Q_1) \text{---} E\alpha \text{---} P(l) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_2^\circ) \text{---} \text{thin}(Q_2) \text{---} E\beta \text{---} P(r) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ Exercise 9.5: $P(\text{in } l) \text{thin}(Q_1 + Q_2) = \text{thin}(Q_1) P(\text{in } l)$	

circuit	Hinze–Marsden
$ \begin{array}{c} A \rightarrow \boxed{\text{ran}(V_1)} \rightarrow \boxed{\frac{1}{\beta}} \rightarrow EA \rightarrow \boxed{E(V_1^\circ)} \rightarrow \boxed{\text{thin}(Q_1)} \rightarrow E\alpha \rightarrow \boxed{P(U_1)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B \\ = \\ A \rightarrow \boxed{\text{ran}(V_2)} \rightarrow \boxed{\frac{1}{\beta}} \rightarrow EA \rightarrow \boxed{E(V_2^\circ)} \rightarrow \boxed{\text{thin}(Q_2)} \rightarrow E\beta \rightarrow \boxed{P(U_2)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B \\ P \text{ a relator and } \text{inl}[U_1, U_2] = U_1 \end{array} $	

$ \text{Map}(\text{cost}) \wedge R = \text{cost} \text{ leq } \text{cost}^\circ \wedge h \text{cost} = F(\text{cost}) k \wedge F(\text{leq}) k \leq k \text{ leq} \implies \text{MonotonicAlg}(h, R) $ if R compares two values by comparing their costs under $\leq$, and h then cost equals $F(\text{cost})$ then a k monotonic on $\leq$, then h is monotonic on R	(15.1.1h)
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circuit	Hinze–Marsden
$ \begin{array}{c} FA \rightarrow \boxed{F(R)} \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow C \\ F(R) h \sqsubseteq h R \text{ iff } F(R) h \text{ cost} \sqsubseteq h \text{ cost} \leq: \text{definition of } R \text{ and shunting} \end{array} $	
$ \begin{array}{c} = \\ FA \rightarrow \boxed{F(R)} \rightarrow \boxed{F(\text{cost})} \rightarrow FC \rightarrow \boxed{k} \rightarrow C \\ \text{assumption } h \text{ cost} = F(\text{cost}) k \end{array} $	
$ \begin{array}{c} \sqsubseteq \\ FA \rightarrow \boxed{F(\text{cost})} \rightarrow \boxed{F(\text{leq})} \rightarrow FC \rightarrow \boxed{k} \rightarrow C \\ R \text{ cost} \sqsubseteq \text{cost} \leq, \text{ as } \text{cost} \text{ is a map; functors} \end{array} $	
$ \begin{array}{c} \sqsubseteq \\ FA \rightarrow \boxed{F(\text{cost})} \rightarrow FC \rightarrow \boxed{k} \rightarrow \boxed{\text{leq}} \rightarrow C \\ \text{assumption } F(\leq) k \leq k \leq: k \text{ monotonic on } \leq \end{array} $	
$ \begin{array}{c} = \\ FA \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow \boxed{\text{leq}} \rightarrow C \\ \text{assumption } h \text{ cost} = F(\text{cost}) k \end{array} $	

the condition	how it is discharged
$F(R)h \sqsubseteq hR$ Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\text{cost})k$, $F(\leq)k \sqsubseteq k \sqsubseteq$	monotonicity when the cost is itself a fold with a step k monotonic on $\leq$
$F(Rn(H^\circ H))h \sqsubseteq hR$ Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$, $F(\leq \times 1)k \sqsubseteq k \sqsubseteq$, H° simple;	monotonicity in context : k may also read the input the part was built from
$QF(H)h \sqsubseteq F(H)hR$ at $Q \triangleq F(U, V)$ Proposition 9.4, U, V preorders, $F(U, R)h \sqsubseteq hR$, $VH \sqsubseteq HR$	both conditions at once, split along the two arguments of a bifunctor

$$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \text{ leq } \text{cost}^\circ \wedge \text{cost} = F(\langle \text{cost}, S \rangle) k \wedge F(\text{leq} \times 1) k \sqsubseteq k \text{ leq} \implies F(R \cap S S^\circ) h \sqsubseteq h R$$

improving each part by R within its S -context, then assembling by h , is below hR

circuit	Hinze–Marsden
$FA \text{---} \boxed{F(R \cap S S^\circ)} \text{---} \boxed{h} \text{---} A$	
$\sqsubseteq FA \text{---} \boxed{F(R \cap S S^\circ)} \text{---} \boxed{h} \text{---} A \text{---} \boxed{\text{cost}} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ shunting: cost a map, so $1 \sqsubseteq \text{cost} \text{ cost}^\circ$ — triple-chains	
$= FA \text{---} \boxed{F(\langle \text{cost} \text{ leq}, S \rangle \langle \text{cost}, S \rangle^\circ)} \text{---} \boxed{h} \text{---} A \text{---} \boxed{\text{cost}} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ products: $R \cap S S^\circ = \langle \text{cost} \text{ leq}, S \rangle \langle \text{cost}, S \rangle^\circ$ at $R = \text{cost} \text{ leq } \text{cost}^\circ \text{---} \triangleq \times$	
$= FA \text{---} \boxed{F(X)} \text{---} \boxed{F(\langle \text{cost}, S \rangle)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ assumption on cost : $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$	
$\sqsubseteq FA \text{---} \boxed{F(\langle \text{cost} \text{ leq}, S \rangle)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ S simple, so $\langle \text{cost}, S \rangle^\circ \langle \text{cost}, S \rangle \sqsubseteq 1$	
$= FA \text{---} \boxed{F(\langle \text{cost}, S \rangle)} \text{---} \boxed{F(\text{leq} \times 1)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ products; functors: $\langle \text{cost} \text{ leq}, S \rangle = \langle \text{cost}, S \rangle (\text{leq} \times 1)$ — bdm-prod-laws, relator-laws	
$\sqsubseteq FA \text{---} \boxed{F(\langle \text{cost}, S \rangle)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} \boxed{\text{leq}} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ assumption on k : $F(\text{leq} \times 1) k \sqsubseteq k \text{ leq}$	

circuit	Hinze–Marsden
$= \text{FA} \xrightarrow{\text{h}} \text{R} \xrightarrow{\text{A}}$ assumption on cost read backwards, then $\text{R} = \text{cost} \text{ leq } \text{cost}^\circ$	

$$\mathbf{G(U,R)h \in hR \wedge Vh \in HR \implies G(U,V)G(1,H)h \in G(1,H)hR}$$

(15.1.1k)

thinning the parts by $\mathbf{U}$ in the first argument and by $\mathbf{V}$ in the second ($\mathbf{G(U,V)}$), then solving by $\mathbf{H}$ and assembling by $\mathbf{h}$, gives only what solving and assembling and then improving by $\mathbf{R}$ gives ($\mathbf{G(1,H)hR}$)

circuit	Hinze–Marsden
$\text{G e w} \xrightarrow{\text{G(U,V)}} \text{G(1,H)} \xrightarrow{\text{G e A}} \text{h} \xrightarrow{\text{A}}$	
$= \text{G e w} \xrightarrow{\text{G(U,VH)}} \text{G e A} \xrightarrow{\text{h}} \text{A}$ taking $\mathbf{Q \triangleq G(U,V)}$; bifunctors: $\mathbf{G(U,V)G(1,H) = G(U,VH)}$ — relator-laws	
$\sqsubseteq \text{G e w} \xrightarrow{\text{G(U,HR)}} \text{G e A} \xrightarrow{\text{h}} \text{A}$ assumption on $\mathbf{V}$: $\mathbf{Vh \in HR}$	
$= \text{G e w} \xrightarrow{\text{G(1,H)}} \text{G(U,R)} \xrightarrow{\text{G e A}} \text{h} \xrightarrow{\text{A}}$ bifunctors: $\mathbf{G(U,HR) = G(1,H)G(U,R)}$ — relator-laws	
$\sqsubseteq \text{G e w} \xrightarrow{\text{G(1,H)}} \text{G e A} \xrightarrow{\text{h}} \text{R} \xrightarrow{\text{A}}$ assumption on $\mathbf{h}$: $\mathbf{G(U,R)h \in hR}$	

§15.2. The string edit problem

$\text{Op} ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}, \text{F(A,B)} = 1 + (\text{A} \times \text{B}), \alpha \triangleq [\text{nil}, \text{cons}].$

(15.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [\text{Op}] \rightarrow [\text{Char}] \times [\text{Char}]$, base returning $([], [])$, step $(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$, step $(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$, step $(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$.

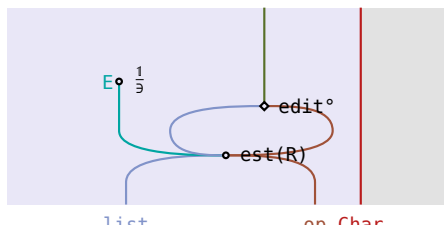
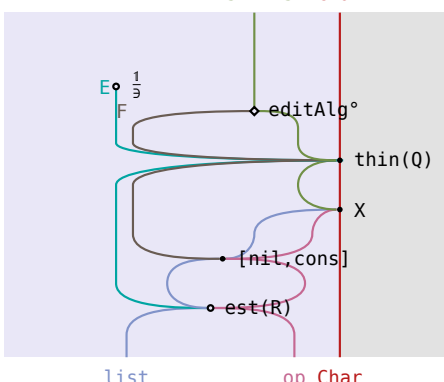
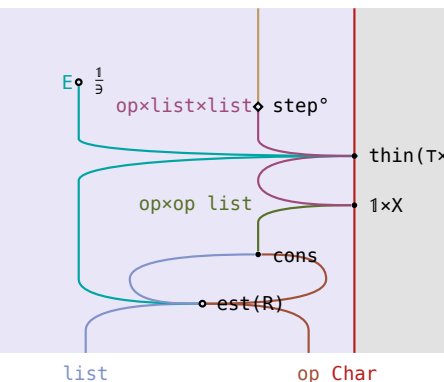
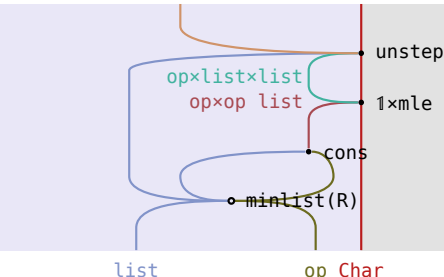
$\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$, $\mathbf{R \triangleq \text{length} \leq \text{length}^\circ}$, $\mathbf{U \triangleq \top}$, $\mathbf{V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ}$, $\mathbf{Q \triangleq 1 + (U \times V)}$, empty the coreflexive on (xs, ys) with both lists empty.

unstep implements $\frac{\text{step}^\circ}{\exists} \text{thin}(U \times V) : \text{unstep } ([a] \# xs, []) = [(\text{del } a, (xs, []))]$, $\text{unstep } ([], [b] \# ys) = [(\text{ins } b, ([], ys))]$, $\text{unstep } ([a] \# xs, [b] \# ys) = (a = b \rightarrow [(\text{cpy } a, (xs, ys))], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$.

$(\mu X : \frac{\text{editAlg}^\circ}{\exists} \text{thin}(Q)P(\text{F}(X)[\text{nil}, \text{cons}])\text{est}(R)) \sqsubseteq \frac{\text{edit}^\circ}{\exists} \text{est}(R)$, $\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle})\text{cons}) \text{minlist}(R))$

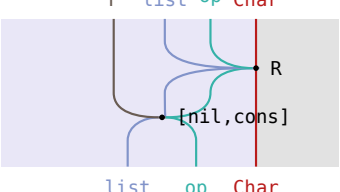
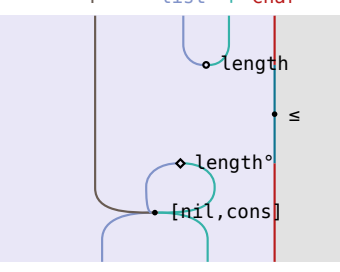
(15.2b)

a shortest edit sequence from which both strings can be reconstituted is one pass over the two of them, each step copying, deleting or inserting one character and the best sequence for what is left taken from the entries already computed

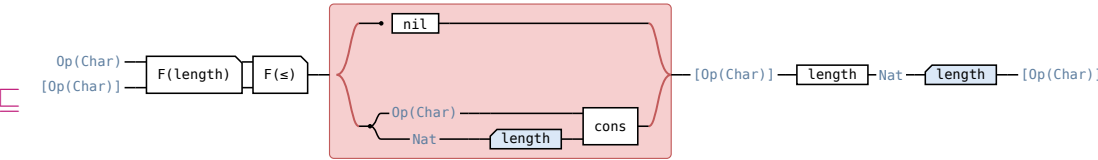
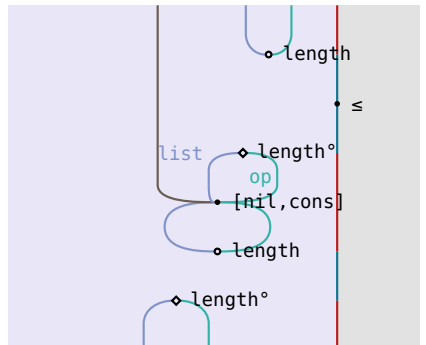
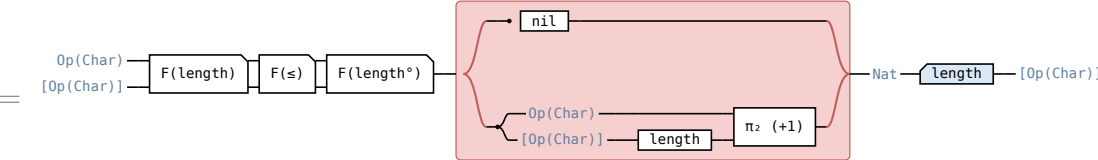
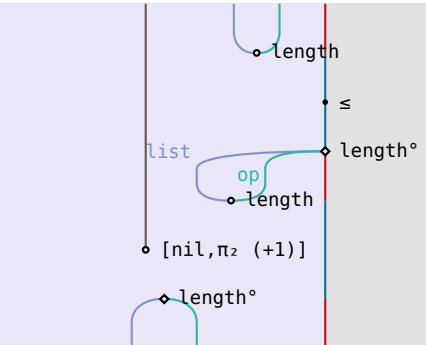
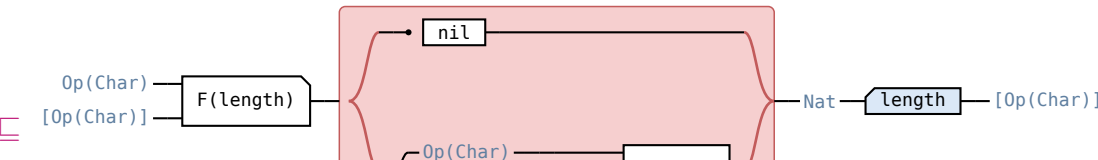
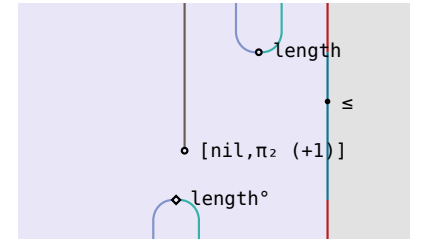
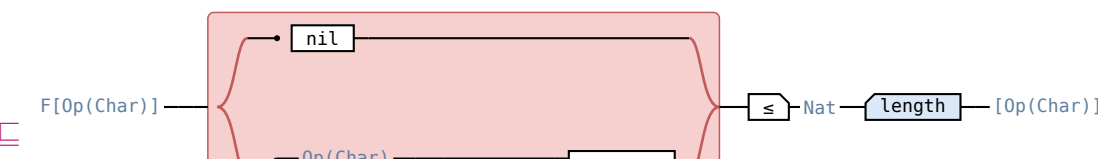
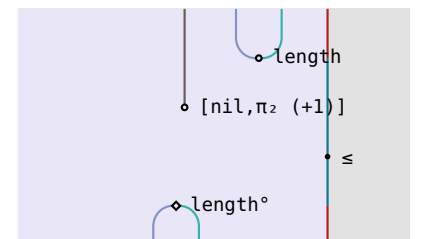
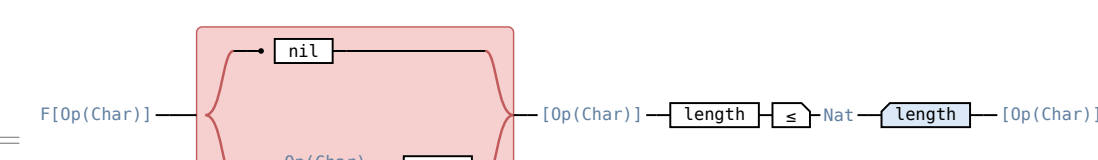
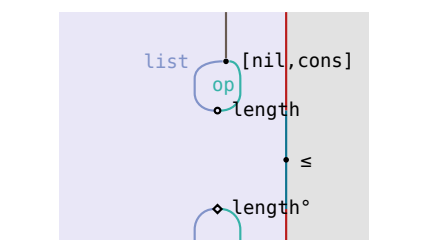
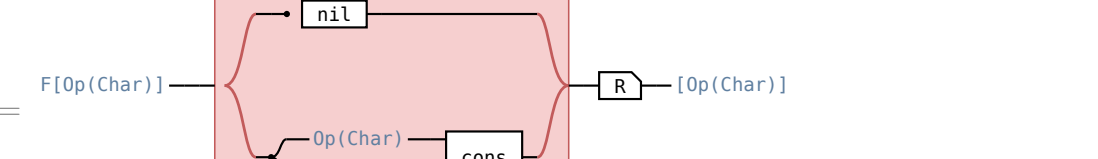
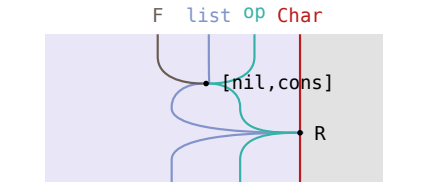
circuit	Hinze–Marsden
$ \begin{array}{c} [\text{Char}] \\ [\text{Char}] \end{array} \xrightarrow{\frac{1}{3}} E[\text{Char}]^2 \xrightarrow{E(\text{edit}^\circ)} E[\text{Op}(\text{Char})] \xrightarrow{\text{est}(\text{R})} [\text{Op}(\text{Char})] $ the specification $\triangleq \text{edit}$	
$ \begin{array}{c} [\text{Char}] \\ [\text{Char}] \end{array} \xrightarrow{\frac{1}{3}} E[\text{Char}]^2 \xrightarrow{E(\text{editAlg}^\circ)} \text{thin}(Q) \xrightarrow{E[\text{Char}]^2} P(F(X)[\text{nil}, \text{cons}]) \xrightarrow{E[\text{Op}(\text{Char})]} \text{est}(\text{R}) \xrightarrow{[\text{Op}(\text{Char})]} $ at $Q \triangleq 1 + (U \times V)$. Monotonicity $F(R) \alpha \sqsubseteq \alpha R$ is Proposition 9.2 at $\text{length} \triangleq ([\text{zero}, \pi_2 \text{succ}])$ with succ monotonic on $\leq$, so cons is monotonic on R. The thinning condition $QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$ over $F(\text{Op}, [\text{Char}] \times [\text{Char}])$ is Proposition 9.4 at $U \triangleq T - F(T, R) \alpha \sqsubseteq \alpha R$, left as an exercise, so any two operations may be compared — and $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$, — $\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}$, $\text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$. $\text{suffix} = \text{tail}^*$ and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ cut those to one step each: drop the operation that produced the head, or weaken its cpy to a del, never lengthening the sequence	
$ \begin{array}{c} [\text{Char}] \\ [\text{Char}] \end{array} \xrightarrow{\frac{1}{3}} E[\text{Char}]^2 \xrightarrow{E(\text{step}^\circ)} \text{thin}(T \times V) \xrightarrow{E(\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}])} P((1 \times X) \text{cons}) \xrightarrow{E[\text{Op}(\text{Char})]} \text{est}(\text{R}) \xrightarrow{[\text{Op}(\text{Char})]} $ $\equiv$ Proposition 9.1: base and step have disjoint ranges, so the fixed point splits into one branch per summand — empty, the coreflexive on (xs, ys) with both lists empty, is where base returns	
$ \begin{array}{c} [\text{Char}] \\ [\text{Char}] \end{array} \xrightarrow{\text{unstep}} [\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}]] \xrightarrow{\text{list}((1 \times \text{mle}) \text{cons})} [[\text{Op}(\text{Char})]] \xrightarrow{\text{minlist}(\text{R})} [\text{Op}(\text{Char})] $ $\equiv$ unstep implements $\frac{\text{step}^\circ}{3} \text{thin}(U \times V)$ — at most two decompositions survive, a copy beating a delete or an insert wherever it is available — and minlist(R) implements est(R). The same subproblem is solved many times over, so the running time is exponential in the two lengths	

$$F(R)[\text{nil}, \text{cons}] \sqsubseteq [\text{nil}, \text{cons}]R$$

comparing the tails by R and then putting an operation in front ($F(R)\alpha$) relates only sequences that R also relates once the operation is in front (αR)

circuit	Hinze–Marsden
$ \begin{array}{c} F[\text{Op}(\text{Char})] \\ \\ \text{Op}(\text{Char}) \\ [\text{Op}(\text{Char})] \end{array} \xrightarrow{\text{nil}} \text{Op}(\text{Char}) \xrightarrow{R} \text{cons} \xrightarrow{[\text{Op}(\text{Char})]} $	
$ \begin{array}{c} \text{Op}(\text{Char}) \\ [\text{Op}(\text{Char})] \end{array} \xrightarrow{F(\text{length})} F(\leq) \xrightarrow{\text{nil}} \text{Op}(\text{Char}) \xrightarrow{\text{length}} \text{cons} \xrightarrow{[\text{Op}(\text{Char})]} $ $R \triangleq \text{length} \leq \text{length}^\circ \triangleq \text{edit}$; F preserves composition	

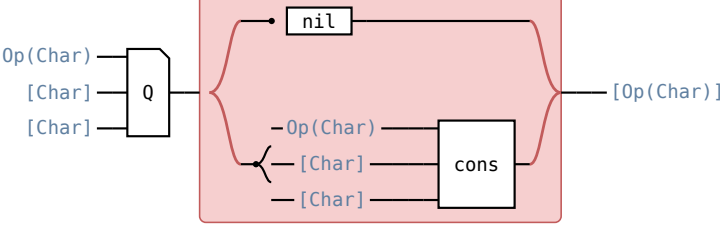
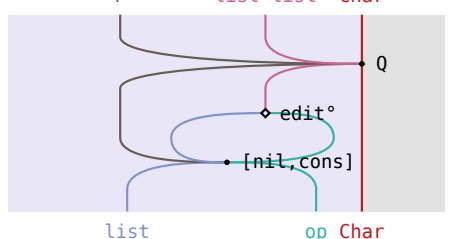
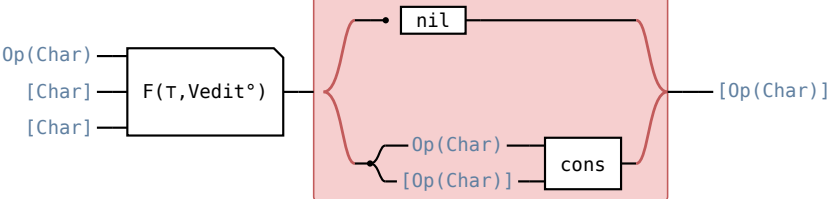
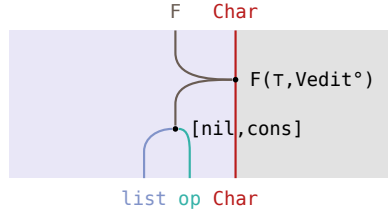
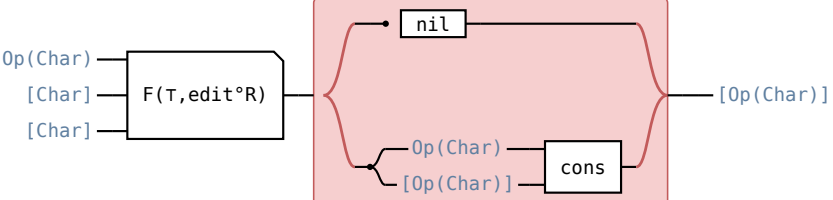
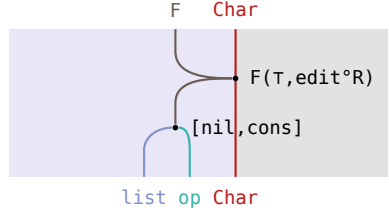
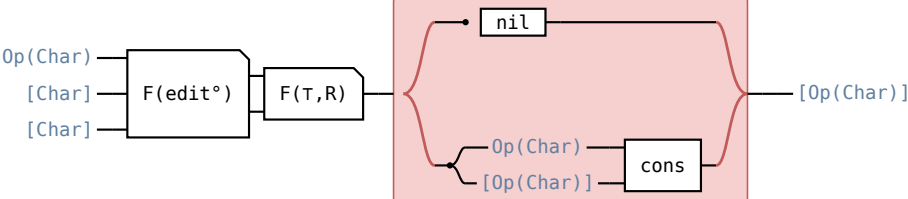
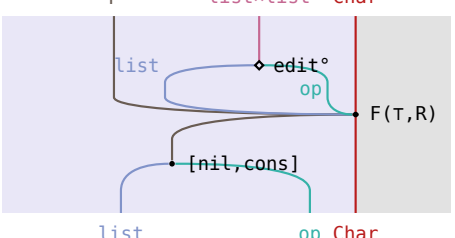
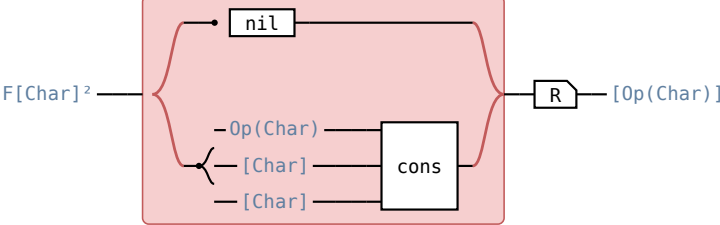
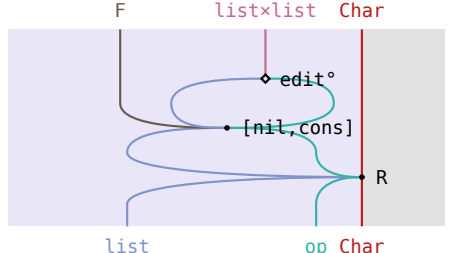
(15.2c)

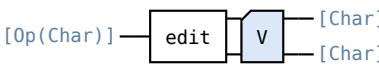
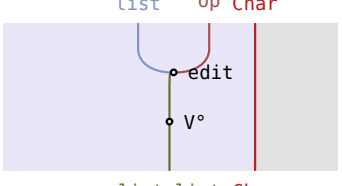
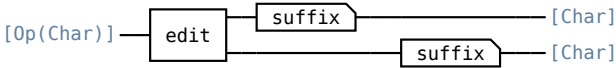
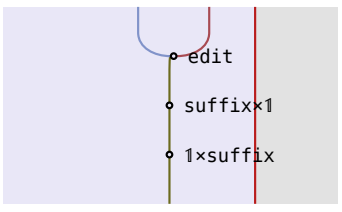
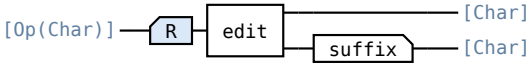
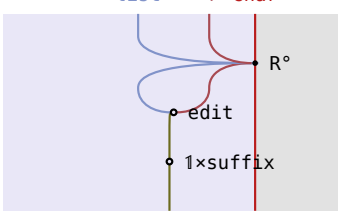
circuit	Hinze–Marsden
 length is a map, so entire: $1 \sqsubseteq \text{length} \text{ length}^\circ$	
 $\alpha \text{ length} = F(\text{length})[\text{zero}, \pi_2 \text{ succ}]: \text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}]) \triangleq \text{edit}$	
 length is a map, so simple, and F is monotonic: $F(\text{length}^\circ)F(\text{length}) \sqsubseteq F(1) = 1$	
 succ is monotonic on $\leq$: $F(\leq)[\text{zero}, \pi_2 \text{ succ}] \sqsubseteq [\text{zero}, \pi_2 \text{ succ}] \leq$	
 $F(\text{length})[\text{zero}, \pi_2 \text{ succ}] = \alpha \text{ length}$ again	
 $R \triangleq \text{length} \leq \text{length}^\circ \triangleq \text{edit}$	

$$QF(\text{edit}^\circ)[\text{nil}, \text{cons}] \sqsubseteq F(\text{edit}^\circ)[\text{nil}, \text{cons}]R$$

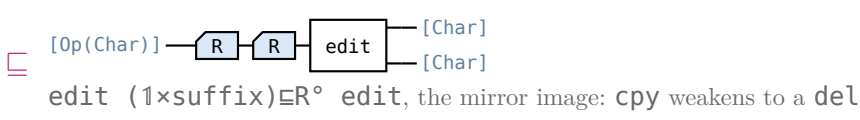
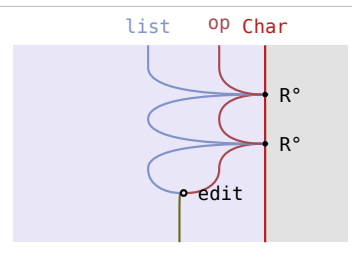
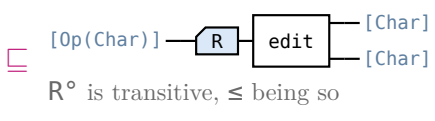
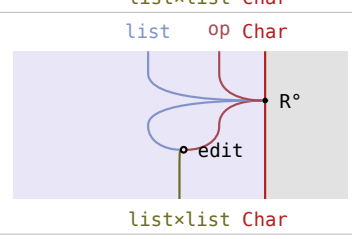
every edit sequence built from a decomposition Q puts above a given one — any operation, each string lengthened at the front — is at least as long as one built from the given one

(15.2d)

circuit	Hinze–Marsden
	
$Q \triangleq 1 + (U \times V)$ is $F(U, V)$ at $U \triangleq \tau \text{ — } \triangleq \text{edit}$; and the bifunctor F preserves composition: $F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$ 	
$\sqsubseteq$ Proposition 9.4's second condition $V \text{ edit}^\circ \sqsubseteq \text{edit}^\circ R \text{ — } (15.2e)$ 	
$=$ F preserves composition: $F(U, \text{edit}^\circ R) = F(1, \text{edit}^\circ)F(U, R)$ 	
$\sqsubseteq$ Proposition 9.4's first condition $F(\tau, R) \alpha \sqsubseteq \alpha R$, left as an exercise in the book: CONS adds one to both lengths whatever the two operations are 	

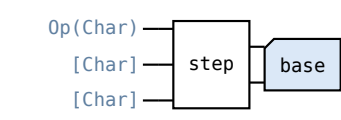
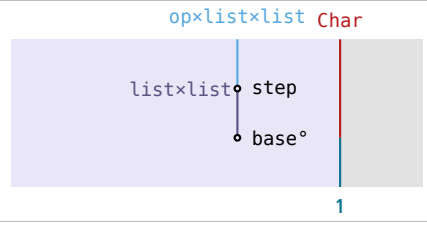
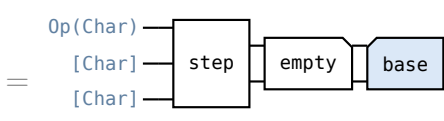
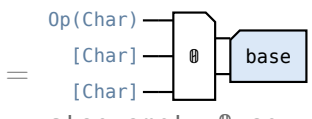
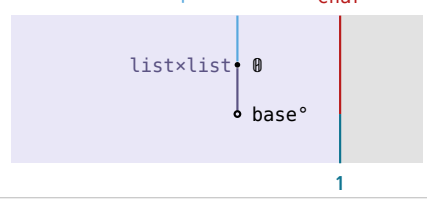
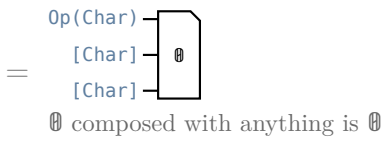
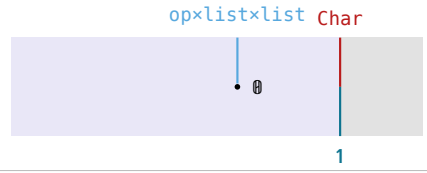
edit $V^\circ \sqsubseteq R^\circ \text{edit}$ cutting a front off either string an edit sequence produces (edit V°) leaves a pair that a sequence no longer produces (R° edit)	
circuit	Hinze–Marsden
	
$=$ $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$, and $\times$ preserves composition 	
$\sqsubseteq$ edit (suffix×1) $\sqsubseteq R^\circ$ edit: drop the operation that produced the head, or weaken its cpy to an ins. Proved by induction at suffix directly, not through suffix=tail* and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ as the book does 	

(15.2e)

circuit	Hinze–Marsden
	
	

step $\text{base}^\circ = \emptyset$

no pair of strings is both a result of **step** and the result $([], [])$ of **base**

circuit	Hinze–Marsden
	
 base returns only $([], [])$, so $\text{base}^\circ = \text{empty } \text{base}^\circ$	
 $\text{step } \text{empty} = \emptyset$: <i>cpy</i> and <i>del</i> put a character on the left string, <i>ins</i> one on the right	
 $\emptyset$ composed with anything is $\emptyset$	

(15.2f)

(15.2g)

$\text{mle}(xs, ys) = \text{head}(\text{column}(xs, ys))$, $\text{column}(xs, ys) = [\text{mle}(u, ys) \mid u \leftarrow \text{tails}(xs)]$ $= \text{column}(xs) = ([\text{fstcol}(xs), \text{nextcol}(xs)] \Downarrow, \text{fstcol} = \text{list}(\text{del}) \text{ tails}$ the tabulation: $\text{mle}(xs, ys)$ needs $\text{mle}(u, v)$ for every tail u of xs and v of ys , so the columns are built right to left	(15.2g)
$\text{column}(xs, \text{cons}(b, ys)) = \text{nextcol}(xs, (b, \text{column}(xs, ys)))$ $= \text{nextcol}(xs)(b, us) = ([\text{base}(b, \text{last}(us)), \text{step}(b)] \Downarrow)(xus)$, $xus = \text{zip}(xs, \text{zip}(\text{init}(us), \text{tail}(us)))$ each column is a fold built bottom to top, over xs zipped with the adjacent pairs of the column to its right	
$\text{base}(b, u) = [[\text{ins}(b)] \# u]$ $= \text{step}(b)((a, (u, v)), ws) = (a \rightarrow b \rightarrow [[\text{cpy}(a)] \# v] \# ws, [\text{bmin}(R)([\text{del}(a)] \# w, [\text{ins}(b)] \# u)] \# ws)$ $w = \text{head}(ws)$; these base , step are not edit 's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths	

§15.3. Optimal bracketing

tree $A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, $\text{FX} = A + X^2$, so $F(R) = 1 + R^2$; $h \triangleq [\text{tip}, \text{bin}]$, $\text{flatten} \triangleq ([\text{wrap}, \text{cat}] \Downarrow) : \text{tree } A \rightarrow \text{list}^+ A$, $H = \text{flatten}^\circ$. $\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}] \Downarrow)$, $\text{opt} \triangleq (\text{zero}, \text{st})$, $\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))$. sb associative, so $\text{size} = \text{flatten } \text{sz}$ for a map $\text{sz} : R \triangleq \text{cost} \leq \text{cost}^\circ$, $g \triangleq [\text{zero}, (1 \times \text{sz})^2 \text{ opb } \pi_1]$, single the coreflexive on singleton lists.	(15.3a)
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$\text{split} \triangleq (\text{inits}^+, \text{tails}^+) \text{ zip}$, an implementation of $\frac{\text{cat}^0}{\text{}};$ $\text{array} \triangleq \text{inits list}(\text{row})$, $\text{row} \triangleq \text{tails list}(\text{mct})$, $\text{col} \triangleq \text{inits list}(\text{mct})$.

$\text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(\text{R})$, $\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}$, $\text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})$.

(15.3b)

$\text{Assoc}(\text{sb}) \Rightarrow F(\langle \text{cost}, \text{flatten} \rangle)g = [\text{tip}, \text{bin}] \text{cost}$ pairing each subtree with its cost and its flattening and then applying g gives what building the tree by $[\text{tip}, \text{bin}]$ and taking its cost gives	
circuit	Hinze–Marsden
$F(\text{tree}(A)) \rightarrow F(\langle \text{cost}, \text{flatten} \rangle) \rightarrow F(\text{Int} \times \text{list}^+(A)) \rightarrow g \rightarrow \text{Int}$	
$=$ definition of g; coproducts and products $\triangleq \text{flatten}$	
$=$ $\text{flatten sz} = \text{size}$, since sb is associative $\triangleq \text{flatten}$	
$=$ $\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])$, at a node $\text{bin}(\text{cost}, \text{size}) = (\text{cost}, \text{size})^2 \text{ opb}$	
$=$ $\text{tip cost} = \text{zero}$; products, $\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}$	
$=$ coproducts, $[\text{tip cost}, \text{bin cost}] = [\text{tip}, \text{bin}] \text{cost}$	

(15.3c)

$F(\leq \times 1)g \leq g$ raising the cost of either subtree ($\leq \times 1$ in both slots) and then applying g returns a value at least what g returns on the costs before raising ($g \leq$)	
circuit	Hinze–Marsden
$F(\text{Int} \times \text{list}^+(A)) \rightarrow F(\leq \times 1) \rightarrow g \rightarrow \text{Int}$	

circuit	Hinze–Marsden
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ definition of $g \triangleq \text{flatten}$	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ definition of opb, and $+$ monotonic	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ $\leq$ reflexive, so $\text{zero} \leq \text{zero} \leq$; coproducts	
$F(\text{Int} \times \text{list}^+(A)) \rightarrow \text{Int}$ definition of $g \triangleq \text{flatten}$	

(15.3d)

$\text{Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\exists} P([\text{tip}, (X \times X) \text{bin}]) \text{est}(R)) \stackrel{\text{flatten}^\circ}{\exists} \text{est}(R), \text{ mct} = (\text{single-head tip}, (\text{init col}, \text{tail row}) \text{mix})$ a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$ is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it	
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circuit	Hinze–Marsden
$\text{list}^+(A) \rightarrow \frac{1}{\exists} \text{E}(\text{list}^+(A)) \rightarrow \text{E}(\text{flatten}^\circ) \rightarrow \text{E}(\text{tree}(A)) \rightarrow \text{est}(R) \rightarrow \text{tree}(A)$ the specification $\triangleq \text{flatten}$	
$\text{list}^+(A) \rightarrow \frac{1}{\exists} \text{E}(\text{list}^+(A)) \rightarrow \text{E}([\text{wrap}, \text{cat}]^\circ) \rightarrow \text{E}(\text{list}^+(A)) \rightarrow P([\text{tip}, (X \times X) \text{bin}]) \rightarrow \text{E}(\text{tree}(A)) \rightarrow \text{est}(R) \rightarrow \text{tree}(A)$ split the list in every way, bracket both halves, join. The condition is monotonicity in context, $F(Rn(\text{flatten} \text{ flatten}^\circ)) \text{h} \leq \text{h} R$ — only trees with the same flattening are compared — which is Proposition 9.3 at $H^\circ = \text{flatten}$, a map, with (9.5) $[\text{tip}, \text{bin}]$ $\text{cost} = (1 + (\text{cost}, \text{flatten})^2) g$ (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6) $(1 + (\leq \times 1)^2) g \leq g \leq$ (g monotonic on $\leq$ in its two cost arguments)	
$\text{list}^+(A) \rightarrow \frac{1}{\exists} \text{E}(\text{list}^+(A)) \rightarrow \text{E}(\text{cat}^\circ) \rightarrow \text{E}(\text{list}^+(A)^2) \rightarrow P((X \times X) \text{bin}) \rightarrow \text{E}(\text{tree}(A)) \rightarrow \text{est}(R) \rightarrow \text{tree}(A)$ Proposition 9.1: wrap and cat have disjoint ranges, and single is the coreflexive on singleton lists, where wrap returns	
$\text{list}^+(A) \rightarrow \text{splits} \rightarrow [\text{list}^+(A)^2] \rightarrow \text{list}((\text{mct} \times \text{mct}) \text{bin}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)$ $\text{splits} \triangleq (\text{inits}^+, \text{tails}^+)$ zip implements $\frac{\text{cat}^\circ}{\exists}$ and $\text{minlist}(R)$ implements $\text{est}(R)$. Exponential, since the segments of one list overlap	

circuit	Hinze–Marsden
$\text{mct} = (\text{single} \rightarrow \text{head } \text{tip}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ mix}) \quad (9.7)$ = the tabulation: mct $\times S$ is needed for every non-empty segment $\times S$, so the values are held as an array of rows, $\text{array} \triangleq \text{inits } \text{list}(\text{row}), \text{row} \triangleq \text{tails } \text{list}(\text{mct}), \text{col} \triangleq \text{inits } \text{list}(\text{mct}), \text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R)$	
$\text{col} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ next}) \quad (9.8)$ $\text{cons } \text{col} = (1 \times \text{array}) \text{ process} \quad (9.9)$ = $\text{row} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{mct}, \text{tail } \text{row}) \text{ cons}) \quad (9.10)$ a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop, $\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}, \text{process} \triangleq ((\text{tip } \text{wrap}) \times 1) \text{ loop}(\text{next})$	
$\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip } \text{wrap } \text{wrap}$ = $\text{addcol} \triangleq (\pi_1 \text{ tip } \text{wrap}, \text{step}) \text{ cons}, \text{step} \triangleq (\text{process } \text{tail}, \pi_2) \text{ zip } \text{list}(\text{cons})$ the program: one fold building the array column by column, cubic in the length of the input	

(15.3e)

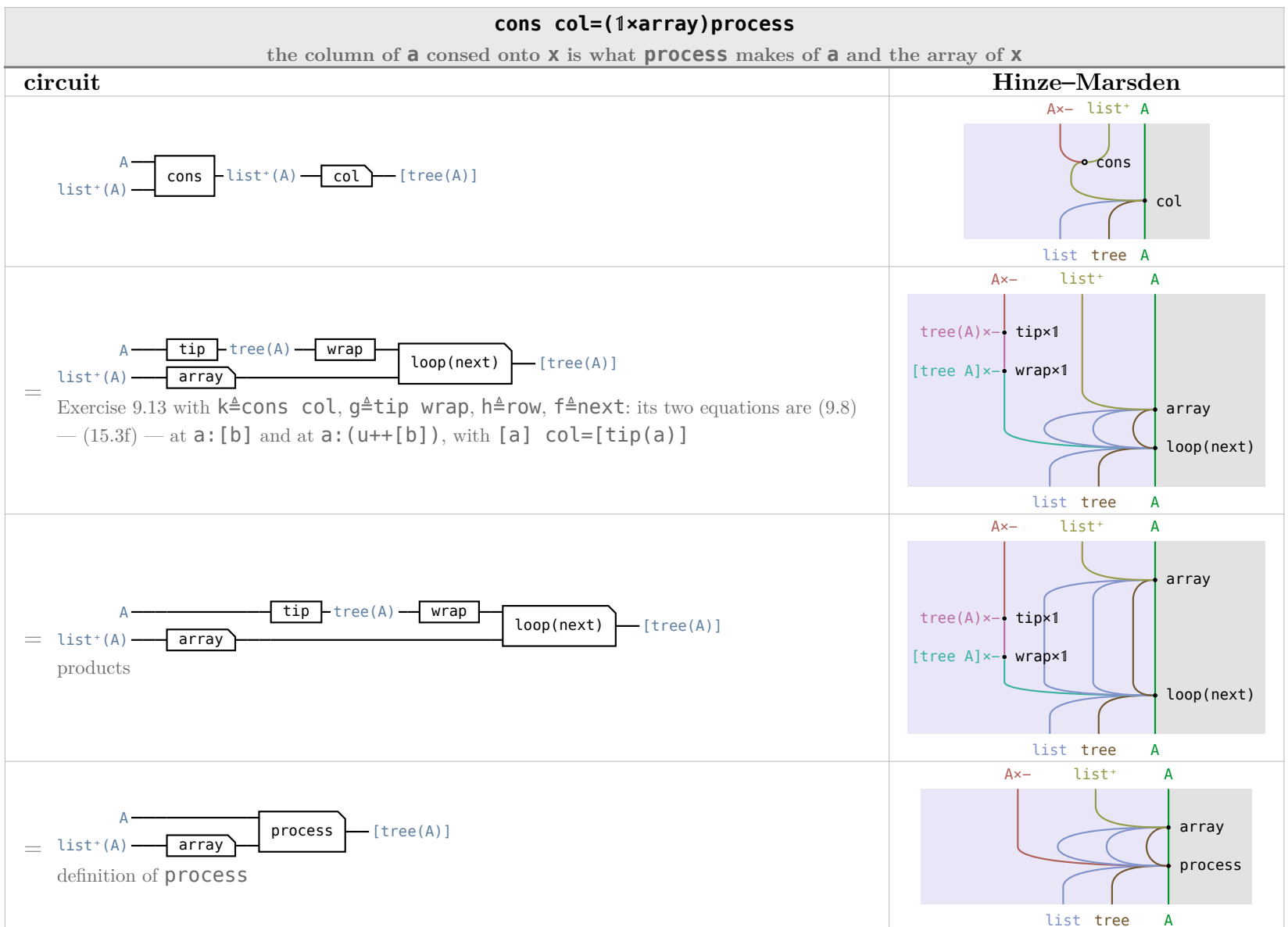
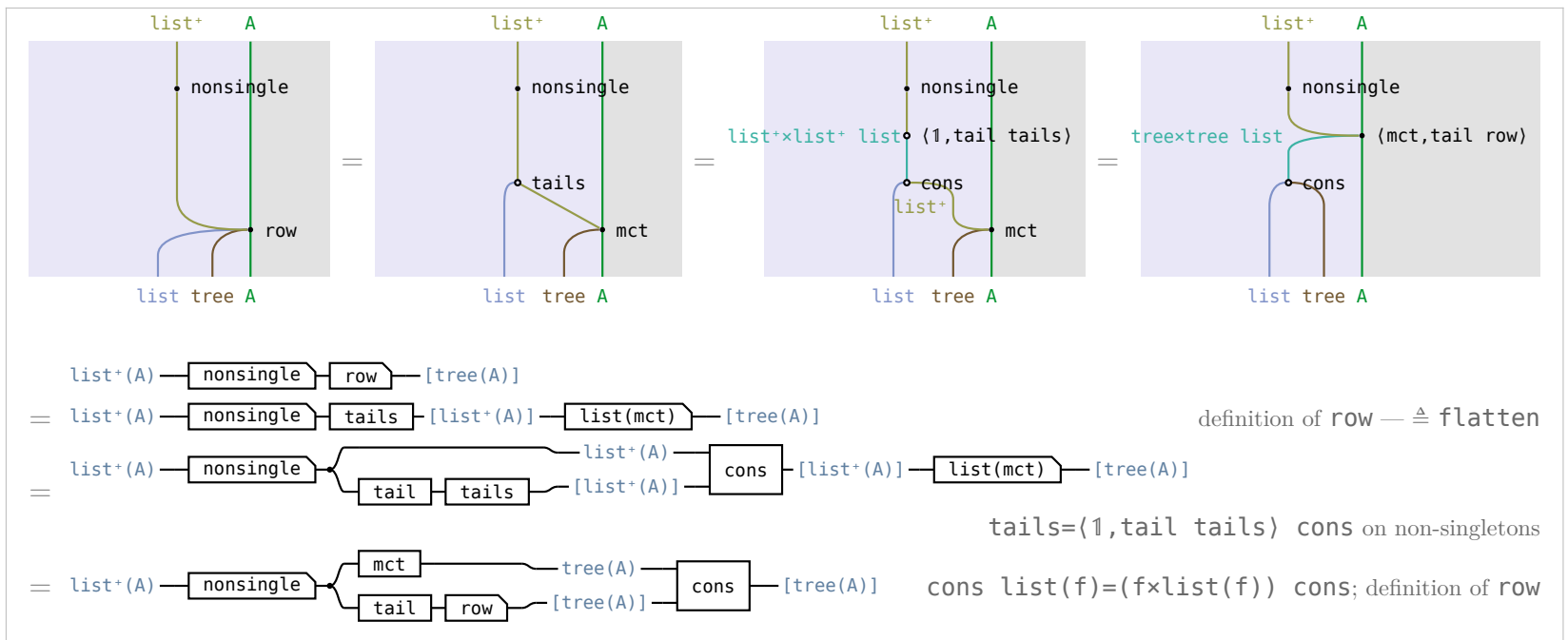
nonsingle mct=nonsingle(init col,tail row)mix on a list of two or more elements, mct gives what mix gives on the column of its init and the row of its tail	
$\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{mct} \rightarrow \text{tree}(A)$ = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{splits} \rightarrow [\text{list}^+(A)^2] \rightarrow \text{list}(\text{graft}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)$ recursive case of $\text{mct} \triangleq \text{flatten}$ = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{inits}^+ \\ \text{tails}^+ \end{cases} \rightarrow \begin{cases} [\text{list}^+(A)] \\ [\text{list}^+(A)] \end{cases} \rightarrow \text{zip} \rightarrow [\text{list}^+(A)^2] \rightarrow \text{list}(\text{graft}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)$ definition of $\text{splits} \triangleq \text{flatten}$ = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{trees} \rightarrow \text{zip} \rightarrow [\text{tree}(A)^2] \rightarrow \text{list}(\text{bin}) \rightarrow [\text{tree}(A)] \rightarrow \text{minlist}(R) \rightarrow \text{tree}(A)$ definitions of $\text{graft} \triangleq (\text{mct} \times \text{mct})$ bin and $\text{trees} \triangleq (\text{inits}^+ \text{ list}(\text{mct}), \text{tails}^+ \text{ list}(\text{mct}))$, our names; $\text{zip } \text{list}(f \times g) = (\text{list}(f) \times \text{list}(g)) \text{ zip}$ = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{trees} \rightarrow \text{mix} \rightarrow \text{tree}(A)$ introducing $\text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R) \triangleq \text{flatten}$ = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} & \text{col} \\ \text{tail} & \text{row} \end{cases} \rightarrow \begin{cases} [\text{tree}(A)] \\ [\text{tree}(A)] \end{cases} \rightarrow \text{mix} \rightarrow \text{tree}(A)$ definition of trees; $\text{inits}^+ = \text{init } \text{inits}$ and $\text{tails}^+ = \text{tail } \text{tails}$ on non-singletons; definition of row and $\text{col} \triangleq \text{flatten}$	

(15.3f)

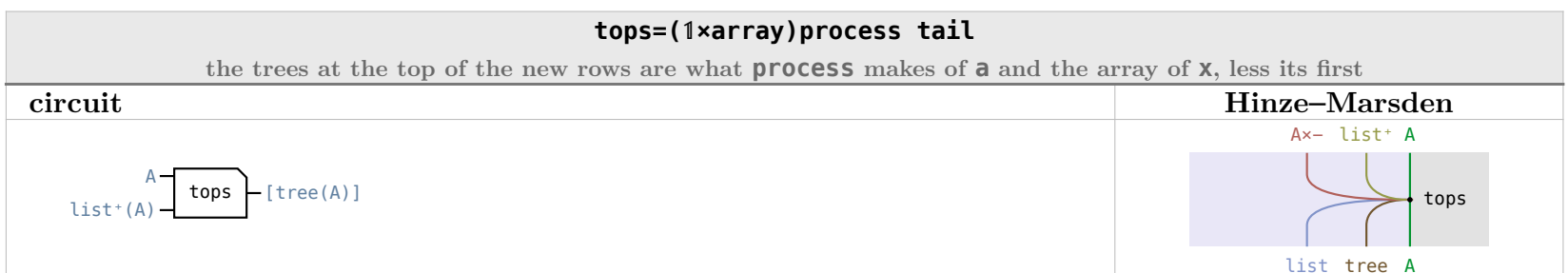
nonsingle col=nonsingle(init col,tail row)next on a list of two or more elements, the column is what next makes of the column of its init and the row of its tail	
$\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{col} \rightarrow [\text{tree}(A)]$ = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \text{inits} \rightarrow [\text{list}^+(A)] \rightarrow \text{list}(\text{mct}) \rightarrow [\text{tree}(A)]$ definition of $\text{col} \triangleq \text{flatten}$ = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} & \text{inits} \\ & \text{list}^+(A) \end{cases} \rightarrow \text{snoc} \rightarrow [\text{list}^+(A)] \rightarrow \text{list}(\text{mct}) \rightarrow [\text{tree}(A)]$ $\text{inits} = (\text{init } \text{inits}, 1)$ snoc on non-singletons = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} & \text{col} \\ \text{mct} & \text{tree}(A) \end{cases} \rightarrow \text{snoc} \rightarrow [\text{tree}(A)]$ $\text{snoc } \text{list}(f) = (\text{list}(f) \times f) \text{ snoc}$; definition of col = $\text{list}^+(A) \rightarrow \text{nonsingle} \rightarrow \begin{cases} \text{init} & \text{col} \\ \text{tail} & \text{row} \end{cases} \rightarrow \begin{cases} [\text{tree}(A)] \\ [\text{tree}(A)] \end{cases} \rightarrow \text{next} \rightarrow [\text{tree}(A)]$ (9.7) on non-singletons — (15.3e) — and introducing $\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc} \triangleq \text{flatten}$	

(15.3g)

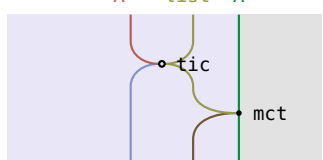
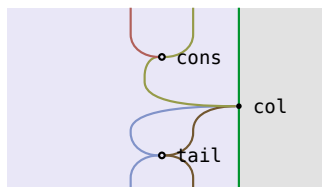
nonsingle row=nonsingle(mct,tail row)cons on a list of two or more elements, the row is mct of the whole list consed onto the row of its tail	
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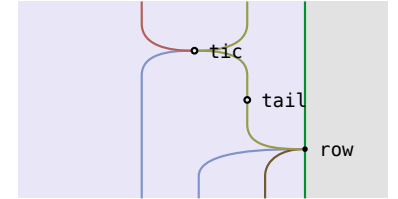
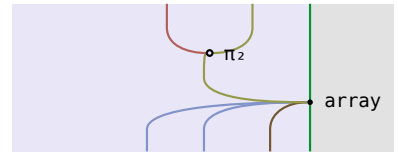


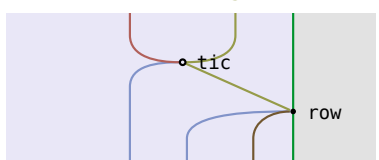
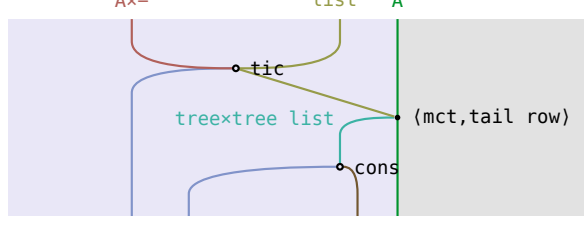
(15.3h)

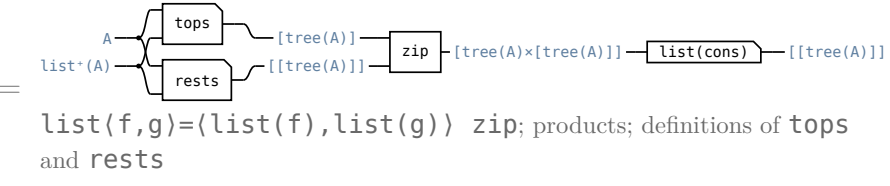
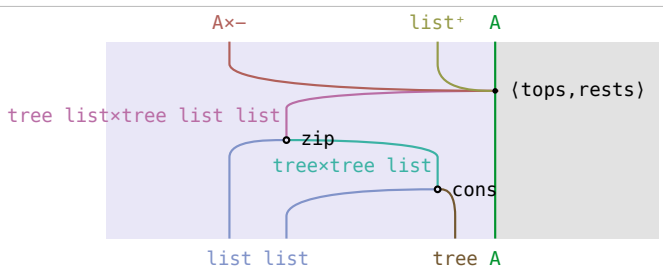
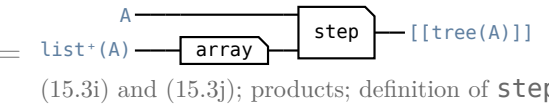


(15.3i)

circuit	Hinze–Marsden
$= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(mct)}} \text{---} [\text{tree}(A)]$ definition of $\text{tops} \triangleq \text{tic}$ list(mct), our name	
$= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{col}} \text{---} \boxed{\text{tail}} \text{---} [\text{tree}(A)]$ $\text{tail list}(f) = \text{list}(f) \text{ tail}$; definition of col	
$= \text{list}^+(A) \text{---} \boxed{\text{array}} \text{---} \boxed{\text{process}} \text{---} \boxed{\text{tail}} \text{---} [\text{tree}(A)]$ (9.9) — (15.3h)	

rests = π_2 array the rest of each new row is the array of $\mathbf{x}$	
circuit	Hinze–Marsden
$\text{list}^+(A) \text{---} \boxed{\text{rests}} \text{---} [[\text{tree}(A)]]$	
$= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(tail row)}} \text{---} [[\text{tree}(A)]]$ definition of $\text{rests} \triangleq \text{tic}$ list(tail row), our name	
$= \text{list}^+(A) \text{---} \text{list}^+(A) \text{---} \boxed{\text{array}} \text{---} [[\text{tree}(A)]]$ $\text{tic list(tail)} = \pi_2 \text{ inits}$; definition of array	

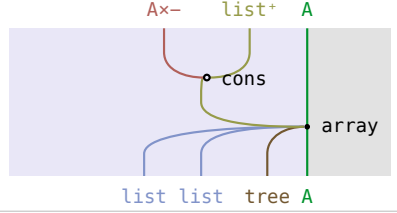
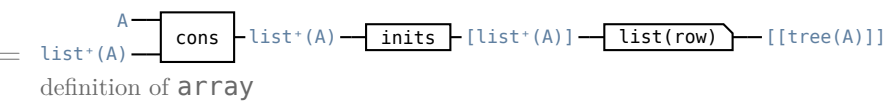
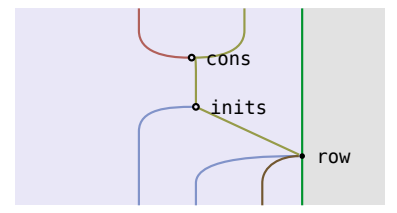
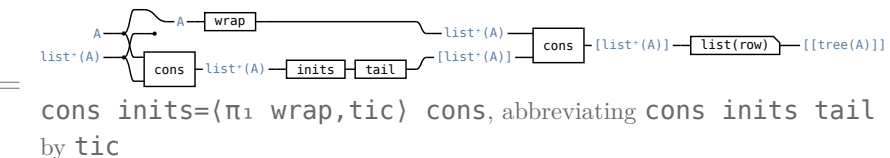
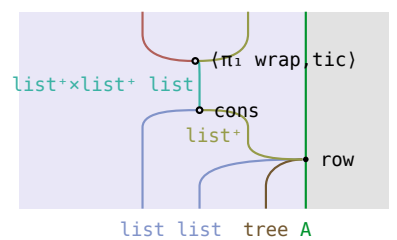
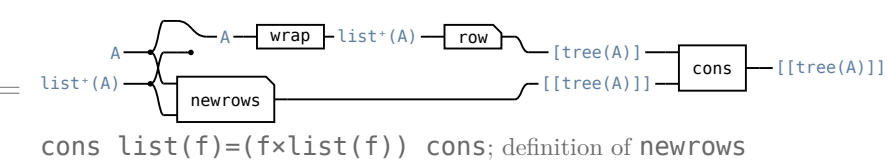
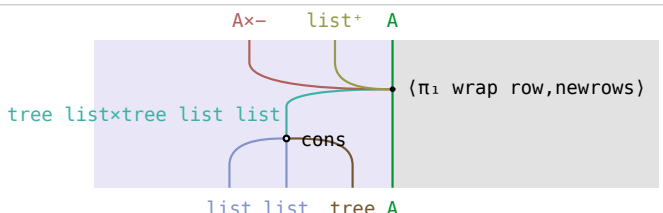
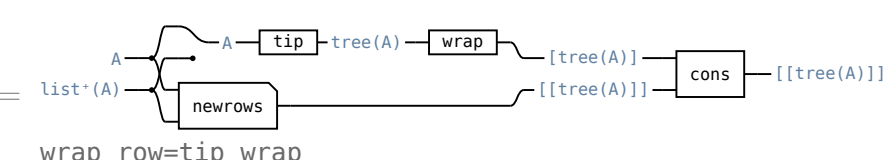
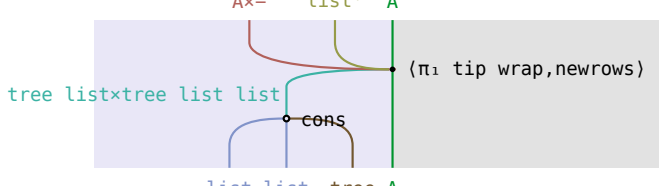
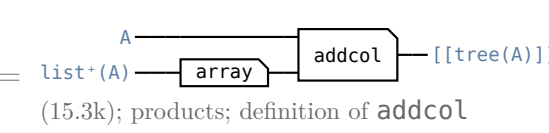
newrows = $(1 \times \text{array}) \text{step}$ the rows of the array of $\mathbf{a}:\mathbf{x}$ below its first are what step makes of $\mathbf{a}$ and the array of $\mathbf{x}$	
circuit	Hinze–Marsden
$\text{list}^+(A) \text{---} \boxed{\text{newrows}} \text{---} [[\text{tree}(A)]]$	
$= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(row)}} \text{---} [[\text{tree}(A)]]$ definition of $\text{newrows} \triangleq \text{tic}$ list(row), our name	
$= \text{list}^+(A) \text{---} \boxed{\text{cons}} \text{---} \text{list}^+(A) \text{---} \boxed{\text{inits}} \text{---} \boxed{\text{tail}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list(mct, tail row)cons}} \text{---} [[\text{tree}(A)]]$ (9.10) — (15.3g) — on the non-singleton tic lists	

circuit	Hinze–Marsden
 $\text{list}(f,g) = (\text{list}(f), \text{list}(g)) \text{ zip}; \text{products}; \text{definitions of tops and rests}$	
 $\text{list}^+(A) \text{ array step } [[\text{tree}(A)]]$ (15.3i) and (15.3j); products; definition of step	

cons array=(1×array)addcol

the array of **a** consed onto **x** is what **addcol** makes of **a** and the array of **x**

(15.3l)

circuit	Hinze–Marsden
 $\text{list}^+(A) \text{ cons array } [[\text{tree}(A)]]$	
 $\text{list}^+(A) \text{ cons inits list(row) } [[\text{tree}(A)]]$ definition of array	
 $\text{list}^+(A) \text{ cons inits tail list(row) } [[\text{tree}(A)]]$ $\text{cons inits} = (\pi_1 \text{ wrap, tic}) \text{ cons}$, abbreviating cons inits tail by tic	
 $\text{list}^+(A) \text{ wrap row cons } [[\text{tree}(A)]]$ $\text{cons list}(f) = (f \times \text{list}(f)) \text{ cons}$; definition of newrows	
 $\text{list}^+(A) \text{ tip wrap cons } [[\text{tree}(A)]]$ $\text{wrap row} = \text{tip wrap}$	
 $\text{list}^+(A) \text{ array addcol } [[\text{tree}(A)]]$ (15.3k); products; definition of addcol	

§15.4. Data compression

$[A] ::= \text{nil} | \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A | \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym Char} | \text{ptr} (\text{String}, \text{String}^+), F(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha^\Delta = [\text{nil}, \text{snoc}].$

(15.4a)

$\text{decode}^\Delta([[\text{nil}, \text{extend}]]): [\text{Code}] \rightarrow \text{String}, \text{ extend } (xs, \text{sym } a) = xs \# [a], \text{ extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; H = \text{decode}^\circ.$

$\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}])$ with c, p the constant costs of a symbol and a pointer; $R \triangleq \text{size} \leq \text{size}^\circ$, $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$, the two τ on symbols and on pointers.

$\text{lrt } ws = \text{est}(\text{prefix}^\circ \times (\tau + \tau)) \{ (xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws \}$, the longest repeated tail; $\text{reduce } (ws \# [a]) = (zs \# [a]) \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)]$ with $(xs, (ys, zs)) = \text{lrt } (ws \# [a])$.

$$0 \leq c \wedge 0 \leq p \implies (\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{3} \text{thin}(Q) P([\text{nil}, (X \times 1) \text{snoc}]) \text{est}(R)) \sqsubseteq \frac{\text{decode}^\circ}{3} \text{est}(R), \text{ encode} = (\text{null} \rightarrow \text{nil}, \text{reduce} \text{list}((\text{encodex1}) \text{snoc}) \text{minlist}(R))$$

(15.4b)

a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

circuit	Hinze–Marsden
the specification — $\triangleq$ decode	
at $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$, the two τ on symbols and on pointers. Monotonicity $F(R) \alpha \sqsubseteq \alpha R$ is routine, the two costs being constants: $(\tau + \tau)[c, p] = [c, p]$. Proposition 9.4 splits the thinning condition in two — $F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$, and $\text{decode prefix} \sqsubseteq R^\circ \text{ decode}$, for which $\text{decode init} \sqsubseteq R^\circ \text{ decode}$ is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins	
Proposition 9.1: nil and extend have disjoint ranges. The decompositions of one string are $\frac{\text{extend}^\circ}{3} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}$ — take the last character as a symbol, or end with a pointer	
reduce implements $\frac{\text{extend}^\circ}{3} \text{thin}(\text{prefix}^\circ \times (\tau + \tau))$: thinning leaves at most two, the symbol and the pointer of the longest repeated tail lrt. Again exponential; the book gives no tabulation for it	

$$0 \leq c \wedge 0 \leq p \implies QF(\text{decode}^\circ)[\text{nil}, \text{snoc}] \sqsubseteq F(\text{decode}^\circ)[\text{nil}, \text{snoc}]R$$

(15.4c)

every code sequence built from a decomposition Q puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one

circuit	Hinze–Marsden

circuit	Hinze–Marsden
$Q \triangleq 1 + (\text{prefix}^\circ \times (T+T))$ is $F(T+T, \text{prefix}^\circ) \triangleq \text{decode}$	
the bifunctor F preserves composition: $F(U, \text{prefix}^\circ) F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ)$	
Proposition 9.4's second condition $\text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R$: dropping the last sym, or shortens or drops the last pointer. Proved at prefix directly by induction, not through init	
F preserves composition: $F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ) F(U, R)$	
Proposition 9.4's first condition $F(T+T, R) \alpha \sqsubseteq \alpha R$, left as an exercise in the book: snoc adds the cost of the last element to both sides, and $[c, p] (T+T) = [c, p]$	

extend nil°=0 no string is both a result of extend and the result [] of nil	
circuit	Hinze–Marsden
nil returns only [], so $\text{nil}^\circ = \text{null } \text{nil}^\circ$	
extend null=0: a symbol ends the string with a character, a pointer with its non-empty zs	

(15.4d)

circuit	Hinze–Marsden
[Char] = Code 0 0 composed with anything is 0	[Char]×- Code 0 1